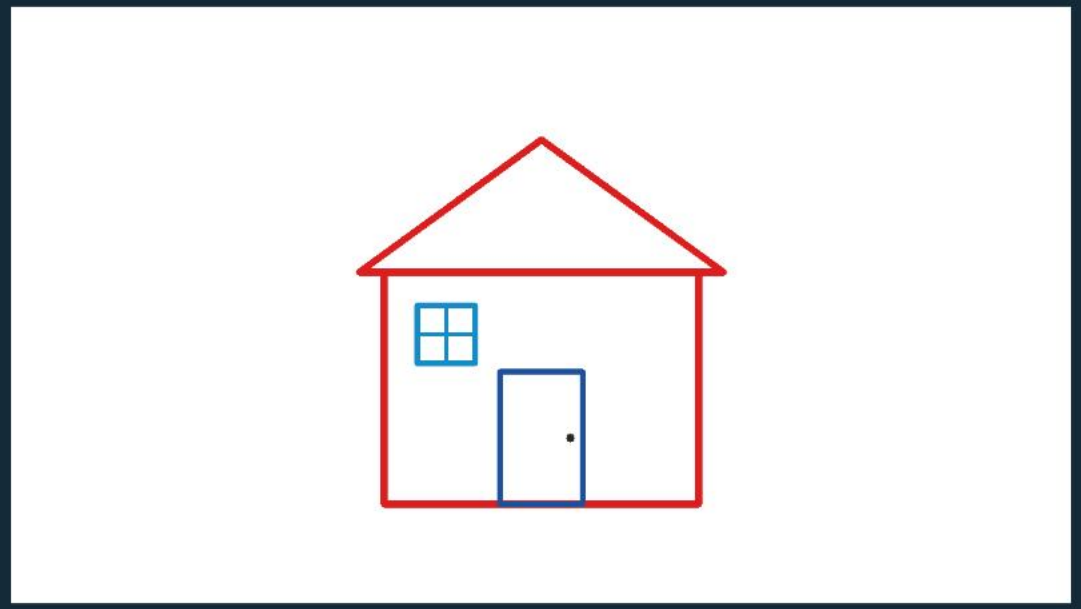
airboard

A whiteboard you draw in the air



Sample board

An AI assisted university project

Who it is for

Students

Sketch a math idea or diagram
while explaining it to someone.

Classroom presenters

Use a laptop webcam for a quick visual explanation.
A drawing tablet is optional.

Drawing gestures

Gesture	Action
Index finger up	**Draw**
Index and middle fingers up	**Erase**
Open hand	**Hover**
Fist held for one second	**Clear the board**

w switches the view. s saves a PNG. q exits.

How drawing works

01	**Webcam**	MediaPipe tracks 21 hand landmarks.
02	**Gesture**	Finger positions choose a drawing mode.
03	**Smoothing**	EMA blends the current and previous position.
04	**Canvas**	OpenCV overlays strokes or shows a white board.

Alpha 1.0 follows raw points. Alpha 0.5 blends them equally.

Notebook pages

Original writing stays beside the text

Multiple pages with automatic saving
Local recognition with Apple Vision
Editable words and PDF export

b: notebook n: new page t: recognize e: edit f: PDF

Handwriting recognition still needs a real writing trial.

How AI helped build it

Antigravity

Created the first implementation and draft materials.

ChatGPT

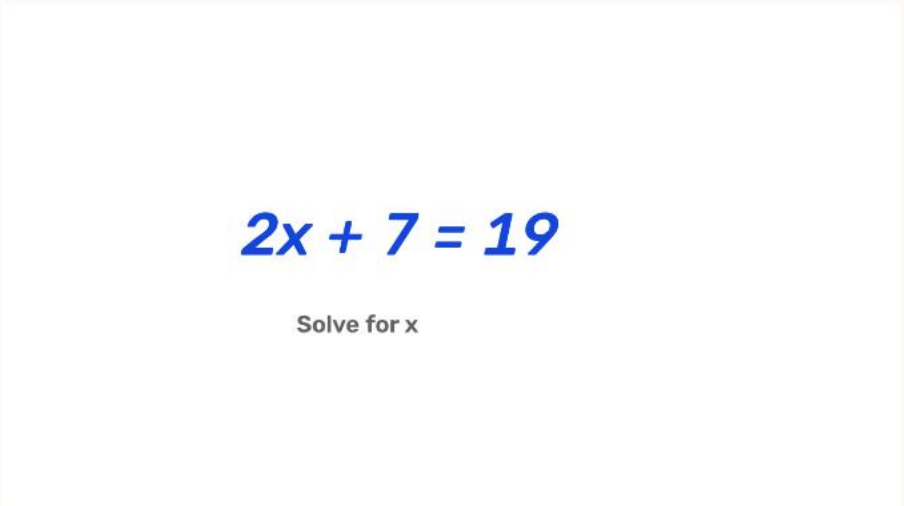
Reviewed the code, reproduced bugs, and helped repair them.

Claude has a planned architecture review role.
The project log records only confirmed contributions.

Optional AI board interpretation

Press a to send the board
to both providers in parallel.

Missing keys and errors stay visible.
Demo mode makes no AI request.



Sample input board

Matching answers can still be wrong.

Live model quality has not been measured.

What we checked

Software checks

Gesture state transitions
PNG export and stroke color
AI failures and comparison logic
Notebook saving and text editing

Still to measure

Real camera accuracy
Lighting and distance behavior
Live AI answer quality
Demo performance on the classroom laptop

Synthetic tests support debugging. Field trials support performance claims.

Class demo and next steps

In class

Draw a short equation or house
Hover, erase, and clear
Switch views and save
Review a notebook page

Next

Record real gesture trials.
Compare smoothing on the same motion.
Grade full AI replies.

Mouse mode is available if the webcam cannot start.